HIGH NORTH ID:
00782084
Date: 2025-09-30
Certificate: 1759247012

High North Inc.
241 Hanlan Rd, Unit 7
Woodbridge, ON, L4L 3R7
1-416-864-6122
LIC-P4PNJMAC20-2022

Client:

Name:

Product:	PAPYB
Lot:	STM-PPB-2508
Matrix:	Flower
Sub-matrix:	Dried Flower
Sampled:	2025-09-17
Received:	2025-09-19

Temperature on Receipt: 22.0°C

Certificate of Analysis

Anhydrous Cannabinoid Analysis	LOD (%)	LOQ (%)	wt%	mg/g
Total THC [(THCA x 0.877) + D9-THC]			29.2217	292.2171
Total CBD [(CBDA x 0.877) + CBD]			0.0967	0.9676
THCA-A	0.03	0.06	32.4438	324.4380
CBGA	0.03	0.06	1.4121	14.1209
D9-THC	0.03	0.06	0.7685	7.6850
CBCA	0.03	0.06	0.2777	2.7773
THCVA	0.03	0.06	0.1947	1.9466
CBG	0.03	0.06	0.1205	1.2048
CBDA	0.03	0.06	0.1103	1.1033
CBC	0.03	0.06	ND	ND
D8-THC	0.03	0.06	ND	ND
CBCVA	0.03	0.06	ND	ND
CBN	0.03	0.06	ND	ND
CBCV	0.03	0.06	ND	ND
THCV	0.03	0.06	ND	ND
CBD	0.03	0.06	ND	ND
CBDV	0.03	0.06	ND	ND
CBDVA	0.03	0.06	ND	ND
Total of all quantified cannabinoids:			35.3276	353.2759

Loss on Drying (EP 2.2.32 Vacuum Oven) used for Anhydrous Cannabinoids Calculation
Method: LAB-MTD-020 (ISO 17025:2017 Accredited)

Moisture Analysis	Result
Loss on Drying (EP 2.2.32 Vacuum Oven)	7.2660%

Method: LAB-MTD-053

Abbreviations: wt% = percentage of weight, CFU = colony forming units, ppm = Parts per million, ppb = Parts per billion, ND = None Detected, BLQ = Below Limit of Quantification, LOQ = Limit of Quantification, LOD = Limit of Detection, RL = Reporting Limit, * = Mixture of Isomers

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Cannabinoid Analysis	LOD (%)	LOQ (%)	wt%	mg/g
Total THC [(THCA x 0.877) + D9-THC]			27.0985	270.9847
Total CBD [(CBDA x 0.877) + CBD]			0.0897	0.8973
THCA-A	0.03	0.06	30.0864	300.8644
CBGA	0.03	0.06	1.3095	13.0949
D9-THC	0.03	0.06	0.7127	7.1266
CBCA	0.03	0.06	0.2575	2.5755
THCVA	0.03	0.06	0.1805	1.8052
CBG	0.03	0.06	0.1117	1.1172
CBDA	0.03	0.06	0.1023	1.0231
CBC	0.03	0.06	ND	ND
D8-THC	0.03	0.06	ND	ND
CBCVA	0.03	0.06	ND	ND
CBN	0.03	0.06	ND	ND
CBCV	0.03	0.06	ND	ND
THCV	0.03	0.06	ND	ND
CBD	0.03	0.06	ND	ND
CBDV	0.03	0.06	ND	ND
CBDVA	0.03	0.06	ND	ND
Total of all quantified cannabinoids:			32.7606	327.6069

Method: LAB-MTD-020 (ISO 17025:2017 Accredited)

Terpene Analysis	LOD (%)	LOQ (%)	wt%
Beta-Myrcene	0.0004	0.005	0.7648
Trans-Caryophyllene	0.0011	0.005	0.5348
(R)-(+)-Limonene	0.0006	0.005	0.2997
Farnesene*	0.0029	0.010	0.2060
Alpha-Humulene	0.0002	0.005	0.1688
Linalool	0.0006	0.005	0.1395
Guaiol	0.0013	0.005	0.1259
Beta-Pinene	0.0004	0.005	0.0515
(R)-Endo-(+)-Fenchyl Alcohol	0.0005	0.005	0.0488
Alpha-Terpineol	0.0007	0.005	0.0426
Alpha-Pinene	0.0002	0.005	0.0380
trans-Nerolidol	0.0005	0.005	0.0167
Caryophyllene oxide	0.0009	0.005	0.0141
Camphene	0.0009	0.005	0.0111
Borneol	0.0005	0.005	0.0072
Terpinolene	0.0005	0.005	BLQ
Squalene	0.0015	0.005	ND
Phytol*	0.0030	0.010	ND
Nootkatone	0.0009	0.005	ND
Farnesol*	0.0032	0.010	ND
Alpha-Bisabolol	0.0011	0.005	ND

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SCC Accredited LAB Accrédité CCN

Terpene Analysis	LOD (%)	LOQ (%)	wt%
Phytane	0.0006	0.005	ND
(+)-Cedrol	0.0004	0.005	ND
cis-Nerolidol	0.0012	0.005	ND
Valencene	0.0006	0.005	ND
Eugenol	0.0010	0.005	ND
Alpha-Cedrene	0.0004	0.005	ND
Geranyl acetate	0.0007	0.005	ND
Carvacrol	0.0005	0.005	ND
Thymol	0.0006	0.005	ND
d-Valerolactam (2-piperidone)	0.0015	0.005	ND
(-)-Piperitone	0.0012	0.005	ND
Isobornyl Acetate	0.0005	0.005	ND
Carvone	0.0006	0.005	ND
Pulegone	0.0006	0.005	ND
Verbenone	0.0006	0.005	ND
Citral*	0.0015	0.005	ND
Geraniol	0.0005	0.005	ND
Safranal	0.0004	0.005	ND
Nerol	0.0007	0.005	ND
Citronellol	0.0008	0.005	ND
Octyl Acetate	0.0005	0.005	ND
Terpinen-4-ol	0.0017	0.005	ND
Camphor	0.0005	0.005	ND
Isoborneol	0.0005	0.005	ND
Menthol (Hexahydrothymol)	0.0013	0.005	ND
Menthone*	0.0015	0.005	ND
Isopulegol	0.0010	0.005	ND
Alpha-Thujone	0.0010	0.005	ND
Fenchone	0.0003	0.005	ND
Sabinene Hydrate	0.0006	0.005	ND
Gamma-Terpinene	0.0002	0.005	ND
Eucalyptol	0.0011	0.005	ND
Cymene*	0.0004	0.005	ND
Ocimene	0.0017	0.005	ND
Alpha-Terpinene	0.0004	0.005	ND
Alpha-Phellandrene	0.0010	0.005	ND
(1S)-3-Carene	0.0009	0.005	ND
Sabinene	0.0003	0.005	ND

Total of all quantified terpenes: 2.470

Method: LAB-MTD-044 (ISO 17025:2017 Accredited)

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Visual Inspection/Olfactory	Result
Foreign Matter	None Detected

Method: LAB-MTD-022 (ISO 17025:2017 Accredited)

Total Ash Analysis	Result
Total Ash (EP 2.4.16)	8.7542%

Method: LAB-MTD-043

Mycotoxin Analysis	LOD (ppb)	LOQ (ppb)	RL (ppb)	Result (ppb)	Status
Aflatoxin-B1	0.5000	2	2	ND	PASS
Aflatoxin-B2	0.5000	2		ND	
Aflatoxin-G1	0.3000	2		ND	
Aflatoxin-G2	0.6000	2		ND	
Sum of Aflatoxins:			4	0	PASS
Method: LAB-MTD-010 (ISO 17025:2017 Accredited)					
Ochratoxin-A	5.6000	20	20	ND	PASS

Method: LAB-MTD-010 (ISO 17025:2017 Accredited)

Microbial Analysis	LOD (CFU/g)	RL (CFU/g)	Result (CFU/g)	Status
Total Yeast and Mold Count	10	20	14	PASS
Total Aerobic Count	10	200	< 10	PASS
Salmonella			Absent in 10g	PASS
S.aureus/P.aeruginosa			Absent in 1g	PASS
E.coli			Absent in 1g	PASS
Bile-Tolerant Gram-Negative			Absent in 1g	PASS

Method: MIC-MTD-001 (ISO 17025:2017 Accredited), MIC-MTD-012 (ISO 17025:2017 Accredited)

Heavy Metals Analysis	LOD (mg/kg)	LOQ (mg/kg)	RL (mg/kg)	Result (mg/kg)	Status
Arsenic	0.015	0.20	3.0	BLQ	PASS
Cadmium	0.006	0.05	0.5	0.0940	PASS
Lead	0.017	0.45	5.0	ND	PASS
Mercury	0.010	0.04	0.5	ND	PASS

Method: LAB-MTD-050 (ISO 17025:2017 Accredited)

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EP 2.8.13 Pesticides Analysis	LOD (ppm)	LOQ (ppm)	RL (ppm)	Result (ppm)	Status
Acephate	0.0422	0.10	0.10	ND	PASS
Alachlor	0.0170	0.05	0.05	ND	PASS
Aldrin and Dieldrin (Sum of)	0.0238	0.05	0.05	ND	PASS
Azinphos-ethyl	0.0416	0.10	0.10	ND	PASS
Azinphos-methyl	0.1154	1.00	1.00	ND	PASS
Bromophos-ethyl	0.0241	0.05	0.05	ND	PASS
Bromophos-methyl	0.0195	0.05	0.05	ND	PASS
Brompropylate	0.0874	3.00	3.00	ND	PASS
Chlordane (Sum of cis-, trans- and oxychlordane)	0.0236	0.05	0.05	ND	PASS
Chlorfenvinphos	0.0694	0.50	0.50	ND	PASS
Chlorpyriphos-ethyl	0.0396	0.20	0.20	ND	PASS
Chlorpyriphos-methyl	0.0281	0.10	0.10	ND	PASS
Chlorthal-dimethyl	0.0032	0.01	0.01	ND	PASS
Cyfluthrin (Sum of)	0.0300	0.10	0.10	ND	PASS
Cypermethrin and isomers (Sum of)	0.0632	1.00	1.00	ND	PASS
DDT (Sum of o,p'-DDE, p,p'-DDE, o,p'-DDT, p,p'-DDT, o,p'-TDE and p,p'-TDE)	0.2493	1.00	1.00	ND	PASS
Deltamethrin	0.1299	0.50	0.50	ND	PASS
Diazinon	0.0836	0.50	0.50	ND	PASS
Dichlofluanid	0.0341	0.10	0.10	ND	PASS
Dichlorvos	0.0589	1.00	1.00	ND	PASS
Dicofol	0.1476	0.50	0.50	ND	PASS
Dimethoate and omethoate (Sum of)	0.0416	0.10	0.10	ND	PASS
Dithiocarbamates (Expressed as CS2)	0.1133	2.00	2.00	ND	PASS
Endosulfan (Sum of isomers and endosulfan sulfate)	0.0836	3.00	3.00	ND	PASS
Endrin	0.0113	0.05	0.05	ND	PASS
Ethion	0.0474	2.00	2.00	ND	PASS
Etrimphos	0.0190	0.05	0.05	ND	PASS
Fenchlorophos (Sum of fenchlorophos and fenchlorophos-oxon)	0.0498	0.10	0.10	ND	PASS
Fenitrothion	0.1398	0.50	0.50	ND	PASS
Fenpropathrin	0.0084	0.03	0.03	ND	PASS
Fensulfothion (Sum of fensulfothion, fensulfothion-oxon, fensulfothion-oxonsulfon and fensulfothion-sulfon)	0.0247	0.05	0.05	ND	PASS
Fenthion (Sum of fenthion, fenthion-oxon, fenthion-oxon-sulfon, fenthion-oxon-sulfoxid, fenthion-sulfon and fenthion-sulfoxid)	0.0246	0.05	0.05	ND	PASS

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EP 2.8.13 Pesticides Analysis	LOD (ppm)	LOQ (ppm)	RL (ppm)	Result (ppm)	Status
Fenvalerate	0.1202	1.50	1.50	ND	PASS
Flucytrinate	0.0245	0.05	0.05	ND	PASS
Fonophos	0.0205	0.05	0.05	ND	PASS
Heptachlor (Sum of heptachlor, cis-heptachlorepoxide and trans-heptachlorepoxide)	0.0230	0.05	0.05	ND	PASS
Hexachlorbenzene	0.0204	0.10	0.10	ND	PASS
Hexachlorocyclohexane (Sum of a-,b-, d- and e)	0.1396	0.30	0.30	ND	PASS
Lambda-Cyhalothrin	0.0860	1.00	1.00	ND	PASS
Lindan (gamma-hexachlorocyclohexane)	0.0574	0.60	0.60	ND	PASS
Malathion and Malaoxon (Sum of)	0.1445	1.00	1.00	ND	PASS
Mecarbam	0.0133	0.05	0.05	ND	PASS
Methacriphos	0.0240	0.05	0.05	ND	PASS
Methamidophos	0.0203	0.05	0.05	ND	PASS
Methidathion	0.0273	0.20	0.20	ND	PASS
Methoxychlor	0.0204	0.05	0.05	ND	PASS
Mirex	0.0031	0.01	0.01	ND	PASS
Monocrotophos	0.0438	0.10	0.10	ND	PASS
Parathion-ethyl and Paraoxon-ethyl (Sum of)	0.1292	0.50	0.50	ND	PASS
Parathion-methyl and Paraoxon-methyl (Sum of)	0.0461	0.20	0.20	ND	PASS
Pendimethalin	0.0463	0.50	0.50	ND	PASS
Pentachloranisol	0.0023	0.01	0.01	ND	PASS
Permethrin and isomers (Sum of)	0.0492	1.00	1.00	ND	PASS
Phosalone	0.0324	0.10	0.10	ND	PASS
Phosmet	0.0209	0.05	0.05	ND	PASS
Piperonyl butoxide	0.1260	3.00	3.00	ND	PASS
Pirimiphos-ethyl	0.0237	0.05	0.05	ND	PASS
Pirimiphos-methyl (Sum of pirimiphos-methyl and N-desethyl-pirimiphos-methyl)	0.1332	4.00	4.00	ND	PASS
Procymidone	0.0404	0.10	0.10	ND	PASS
Profenophos	0.0422	0.10	0.10	ND	PASS
Prothiophos	0.0166	0.05	0.05	ND	PASS
Pyrethrum (Sum of cinerin I, cinerin II, jasmolin I, jasmolin II, pyrethrin I and pyrethrin II)	0.1233	3.00	3.00	ND	PASS
Quinalphos	0.0177	0.05	0.05	ND	PASS
Quintozene (Sum of quintozene, pentachloraniline and methyl penthachlorphenyl sulfide)	0.0804	1.00	1.00	ND	PASS
S-421	0.0093	0.02	0.02	ND	PASS
Tau-fluvalinate	0.0181	0.05	0.05	ND	PASS

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EP 2.8.13 Pesticides Analysis	LOD (ppm)	LOQ (ppm)	RL (ppm)	Result (ppm)	Status
Tecnazene	0.0183	0.05	0.05	ND	PASS
Tetradifon	0.1194	0.30	0.30	ND	PASS
Vinclozolin	0.1031	0.40	0.40	ND	PASS

Method: LAB-MTD-041

Identification A (Macroscopic) DAB Monograph

Complies with monograph

Bracts and flowers of the overall inflorescence form the flattened branched raceme in which each branch has more than one flower.
This highly compressed panicle is approximately 1 to 5 cm in length and width.

The flower husks are green to light green, covered with dense yellow-white hairs, and stuck together with resin. The flower is about 5 to 10 mm long, consisting of a hooded, green to light green bloom.

Light brown to brown pistils and stigma branches, within an individual flower, having an overall length of up to 1 cm.

The crumbled inflorescence contains peduncle fragments, bracts, and panicle sections, as well as individual flowers and flower organs.

Bracts and all flower organs, except pistils, are more or less densely covered with excreted resin-adhesive glandular hairs.

Singular brown ovule contained within the base of the flower with two thin stigma protrusions

Method: MIC-MTD-010

Identification B (Microscopic) DAB Monograph

Complies with monograph

Isolated Stalks

Isolated Heads

The upper epidermis having polygonal or sinuate anticlinate cell walls, the cystolith trichomes having very thick, in some cases verrucose cell walls, the cystoliths can be seen as botryoid structures, the palisade parenchyma is visible below the epidermis; bract fragments having fine, unicellular covering trichomes. (Top and Side view)

Method: MIC-MTD-010

Identification C (by TLC) DAB Monograph

CBD absent and THCA present. (Conforms)

Method: LAB-MTD-052

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Comments

Testing has been conducted as per TGO specifications.
This COA cancels and supersedes certificate 1758820322 dated 2025-09-25.

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